HEALTHCARE ANALYTICS CAPSTONE PROJECT

Scalability & Future Plan Report

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Team ID	Individual Project
Project Name	Health360_HealthCare_Analysis
Author / Analyst	Kriti Bhandari
Domain	Healthcare Analytics
Maximum Marks	1 Mark

Scalability & Future Plan

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Flat-File Ingestion Dependency	Relies on static CSV/Excel upload (N=5,000 records) rather than direct live REST API streaming from enterprise EMR/EHR systems.	High
2	Lack of Embedded Machine Learning Engine	Provides descriptive & diagnostic analytics but lacks automated predictive ML scoring (e.g., XGBoost 5-year stroke/heart failure risk).	High
3	Manual Alerting Mechanism	High-risk outliers are identified visually on scatter plots rather than pushing automated real-time SMS/Email alerts to clinical care teams.	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Hosted on Tableau Public for concurrent public/analyst review (~100 daily active users).	Deploy to Tableau Server / Cloud with Role-Based Access Control (RBAC) to support 10,000+ clinicians and policy planners.
2	Data Storage	Snowflake database storing 5,000 patient records (2.4 MB compressed storage).	Scale architecture via dynamic Snowflake auto-clustering to seamlessly handle 500,000+ patient records with sub-second queries.
3	Performance	Query execution latency ranges from 45 ms to 210 ms using XSmall Snowflake warehouse.	Implement Materialized Views and Snowflake Micro-Partitioning to maintain sub-100 ms rendering times at enterprise scale.
4	Security	PII anonymization applied during CSV preprocessing.	Implement end-to-end HIPAA-compliant data encryption (SSL/TLS in transit, AES-256 at rest) with row-level security (RLS).

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Real-Time Pipeline Integration	Q1 2027 (Jan - Mar)	Connect Snowflake directly to live EMR/EHR APIs for automated real-time biometric feed ingestion.
Phase 3	Predictive Risk ML Engine	Q2 2027 (Apr - Jun)	Integrate XGBoost / Random Forest models to calculate individual 5-year chronic disease progression scores.
Phase 4	Automated Clinical Alerting System	Q3 2027 (Jul - Sep)	Deploy automated SMS/Email/EHR push notifications for care teams when blood pressure or glucose exceeds thresholds.